

Parametric Mechanical Components as a Service

Problem

Engineers, makers, robotics teams, maintenance technicians, and small businesses frequently need custom mechanical parts:

- Timing pulleys
- Gears
- Couplers
- Encoder mounts
- Motor mounts
- Adapter plates
- Sensor brackets
- SBC enclosures
- Camera mounts
- Bearing blocks
- Belt tensioners

Most of these parts are simple to design but require 30–120 minutes of CAD work.

The customer is not paying for the CAD. They are paying to avoid the CAD.

Solution

A web application where users enter engineering parameters and immediately receive:

- STL
- STEP
- OpenSCAD
- Fusion 360 archive
- Drawing PDF

Example:

"I have two shafts 72mm apart and an S3M-270 belt."

Output:

- Correct pulley tooth counts
- Shaft bores
- Flanges
- Lightening holes
- Print-ready STL
- BOM

Generated in under one minute.

Target Customers

Hobbyists

- 3D printer owners
- Arduino users
- Raspberry Pi users
- Robotics hobbyists

Professional Users

- Maintenance technicians
- Manufacturing engineers
- Controls engineers
- Integrators
- Prototype shops

Small Businesses

- Product startups
- Machine builders
- Automation companies

Revenue Model

Tier 1

Free STL

Purpose:

- Search engine traffic
- Community growth

Tier 2

\$5-10

Download:

- STEP
- Editable CAD
- Drawings

Tier 3

\$29/month

Unlimited downloads

Tier 4

\$99–299/month

Commercial subscription

Includes:

- API access
- Team accounts
- White-label generation

Distribution

SEO

Pages such as:

- S3M Pulley Generator
- GT2 Pulley Generator
- HTD Belt Calculator
- Spur Gear Generator
- D Shaft Coupler Generator

Each tool becomes a search landing page.

YouTube

Videos:

- Designing a pulley in 30 seconds
- Custom encoder mount generated automatically
- From dimensions to STL in 60 seconds

Hackaday

Build logs demonstrating generated designs.

Reddit

Relevant communities:

- r/3Dprinting
- r/functionalprint
- r/robotics
- r/arduino
- r/raspberry_pi

Competitive Advantage

Most CAD tools are built for CAD users.

This service is built for people who need a part.

The user thinks in:

- Shaft diameter
- Belt type
- Mounting pattern

Not sketches and constraints.

Future Expansion

Phase 1:

- Pulleys
- Gears
- Couplers

Phase 2:

- Brackets
- Enclosures
- Adapter plates

Phase 3:

- Full assemblies

Example:

"Generate a belt reduction from a NEMA17 to an AMT102 encoder."

Output:

- Complete assembly
- BOM
- STL
- STEP
- Drawings

Generated automatically.

Long-Term Vision

An engineer describes a mechanism in plain English.

The platform generates:

- CAD
- Manufacturing files
- Drawings
- BOM

The value is not CAD generation.

The value is converting engineering intent into manufacturable hardware.